Arduino: Ultrasonic Distance Sensor (HC-SR04)

Comprehensive Lesson Guide & Practical Implementation

Real-World Scenario: How Do Cars & Auto-Rickshaws Know When to Stop?
Have you ever seen a car stop just before hitting a wall while parking? Imagine Rahul's uncle parking his auto-rickshaw in a tight Bengaluru lane. How does he avoid bumping into obstacles? Vehicles use proximity sensing technology powered by sound waves!

💡 **Fun Fact:** Bats and dolphins use *echolocation* to 'see' and navigate in the dark using sound waves — exactly like the HC-SR04 sensor!

1. Introduction to the HC-SR04 Sensor

The HC-SR04 is an ultrasonic distance measurement module that provides stable distance readings from 2 cm to 400 cm with high accuracy.

Pin Name	Arduino Connection	Function Description
VCC	5V Pin	Provides 5V power supply to the sensor module.
GND	GND Pin	Ground connection to complete the circuit.
Trig (Trigger)	Digital Pin 2	Input pin on sensor; receives pulse from Arduino to emit sound waves.
Echo	Digital Pin 3	Output pin on sensor; sends pulse to Arduino when reflected sound is received.

2. Physics of Measurement: How It Works

The sensor transmits a short 40 kHz ultrasonic pulse through air. When the wave encounters an obstacle, it bounces back (echoes) to the receiver.

$$\text{Distance} = (\text{Speed of Sound} \times \text{Time}) \div 2$$

Why divide by 2? Sound travels from the transmitter *TO* the object and then reflects *BACK* to the receiver. That is two full trips for one measurement! Dividing by two isolates the one-way distance.

Key Constant: Speed of Sound in Air ≈ **343 m/s** (or **0.034 cm/μs**).

3. Coding Step-by-Step Guidance

Step 1: Set Up Pins

Configure Pin 2 as `OUTPUT` (Trig) and Pin 3 as `INPUT` (Echo) inside your `setup()` block.

Step 2: Send a Sound Pulse

Write `LOW` for 2μs, then `HIGH` on the Trig pin for 10 microseconds to emit an ultrasonic burst, then set it back to `LOW`.

Step 3: Read Distance in cm

Use `pulseIn(echoPin, HIGH)` to measure duration in microseconds. Calculate distance using:

Distance (cm) = (time × 0.034) ÷ 2

Step 4: Serial Monitor Setup

Initialize `Serial.begin(9600)` and use `Serial.print(distance)` to view real-time measurements on screen.

Step 5 & 6: Logic Control (Alert System)

IF distance < 15 cm: Turn ON Red LED (Pin 7) and Buzzer (Pin 8).
ELSE: Turn OFF LED and Buzzer.

Student Assessment & Practice Questions

Test your understanding of the HC-SR04 Ultrasonic Sensor & Arduino

Question 1: **Which pin on the HC-SR04 sensor is responsible for sending out the ultrasonic sound wave pulse?**

A) VCC B) Trig C) Echo D) GND

Question 2: **Why is the total measured travel time of the ultrasonic wave divided by 2 in the distance calculation formula?**

A) To convert microseconds into milliseconds B) Because sound travels at twice the normal speed in air C) Because the wave travels to the obstacle and back, covering double the distance D) To account for the voltage drop across the sensor pins

Question 3: **Ultrasonic sound waves operate in a frequency range that is:**

A) Directly audible to human ears B) Below 20 Hz (Infrasonic) C) Above 20 kHz (Inaudible to humans) D) Identical to FM radio frequency waves

Question 4: **If the `pulseIn()` function records a travel time of 1000 microseconds, what is the calculated distance to the object using** $speed = 0.034\ cm/\mu s$**?**

A) 34 cm B) 17 cm C) 68 cm D) 8.5 cm

Question 5: **In an automated obstruction detection code, if an object is detected at 10 cm and the threshold condition is set to `if (distance < 15)`, what action will the system execute?**

A) Keep the alert LED and buzzer deactivated B) Activate the alert system (turn ON LED and Buzzer) C) Reset the Arduino controller D) Disconnect power to the VCC pin

Question 6: **What mode must the Arduino pin connected to the sensor's Echo pin be set to using the `pinMode()` function?**

A) OUTPUT B) INPUT C) HIGH D) LOW

🌧️ Mini Challenge: Design Task

Scenario: You are tasked with designing a *Monsoon Flood Early Warning System* using the Arduino and HC-SR04 ultrasonic sensor.

Your Task: Write out the circuit diagram layout and logical flow (pseudocode or block plan) explaining how you would position the sensor above a river or drainage canal to detect rising water levels and trigger emergency alerts at different critical height thresholds.